A DECENT SPACE COVERED ETALE-SURJECTIVELY BY A FIELD SPECTRUM IS THE SPECTRUM OF A FINITE SEPARABLE EXTENSION

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Editorial note. Counted span is the author's own proof environment, source lines 3026-3035, immediately following the statement at 3018-3024.

Editorial note. The closing 'and we win' at source line 3035 compresses two facts (k'/k finite separable from etaleness; a monomorphism that is etale surjective is an isomorphism); the expansion recorded in proof_roadmap is the reviewer's checklist, not author text.

Editorial note. One cross-chapter pointer (Schemes) is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Decent points are represented by quasi-compact monomorphisms. decency property invoked at proof lines 3030-3031 (find a quasi-compact monomorphism Spec(k') -> X with image x). Produces the candidate isomorphism Spec(k') -> X from decency of X.
- Monomorphisms towards a field spectrum. Schemes, Lemma schemes-lemma-mono-towards-spec-field, cited at proof lines 3033-3034. U -> Spec(k) is a monomorphism of schemes, hence U = Spec(k) (or empty; nonempty since it receives the point).

Source reference. Original reference 03DP: focused-seeds/geometry/S9-decent-decent-point-like-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let k be a field. Let X be an algebraic space over S and assume that there exists a surjective étale morphism $\mathrm{Spec}(k) \to X$. If X is decent, then $X \cong \mathrm{Spec}(k')$ where k/k' is a finite separable extension.*

Proof. The assumption implies that $|X| = \{x\}$ is a singleton. Since X is decent we can find a quasi-compact monomorphism $\mathrm{Spec}(k') \to X$ whose image is x. Then the projection $U = \mathrm{Spec}(k') \times_X \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is a monomorphism, whence $U = \mathrm{Spec}(k)$, see Schemes, Lemma 03DP. Hence the projection $\mathrm{Spec}(k) = U \to \mathrm{Spec}(k')$ is étale and we win. □

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